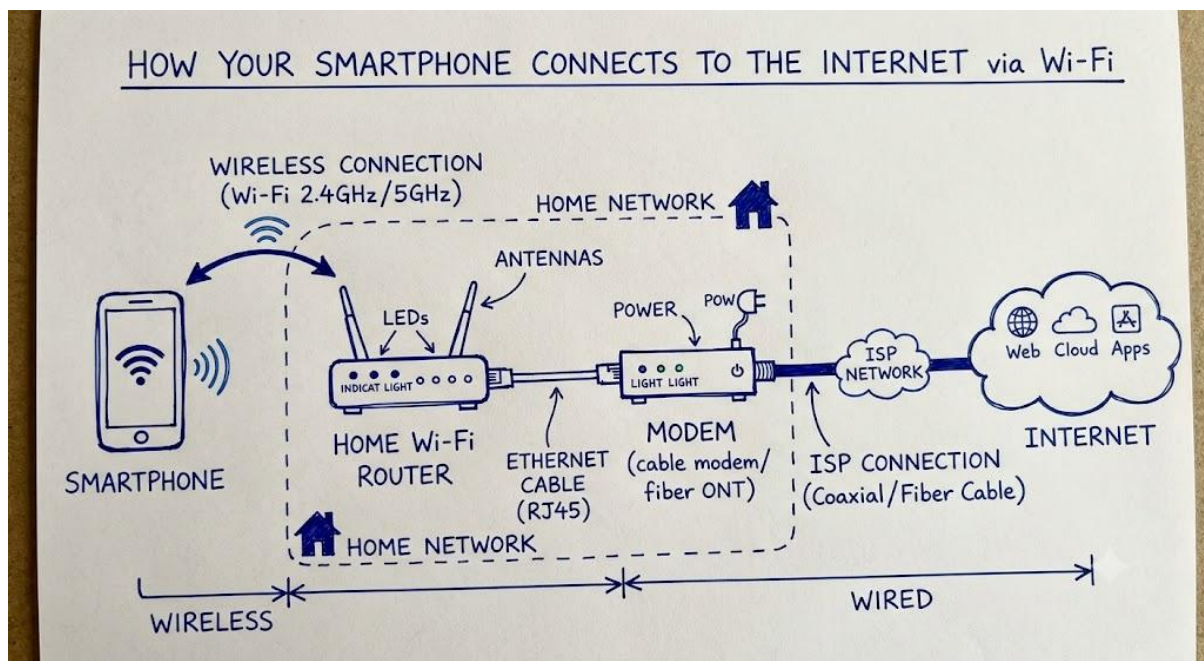


TASK-1

Wi-Fi Standard	Frequency Band	Maximum Theoretical Speed	Key Difference
802.11n (Wi-Fi 4)	2.4 GHz and 5 GHz	Up to 600 Mbps	First standard to support both 2.4 GHz and 5 GHz bands with improved speed using MIMO technology.
802.11ac (Wi-Fi 5)	5 GHz only	Up to 3.5 Gbps	Operates only on the 5 GHz band , providing much faster speeds and better performance than 802.11n.
802.11ax (Wi-Fi 6)	2.4 GHz and 5 GHz (Wi-Fi 6E also adds 6 GHz)	Up to 9.6 Gbps	Offers the highest speed and better performance in crowded networks using technologies such as OFDMA and improved MU-MIMO.

TASK-2



TASK-3

SSID Name)	(Wi-Fi Signal Strength	Security Type	Observation
Home_WiFi_5G	Excellent (-42 dBm)	WPA3-Personal	Strong signal, latest security, best choice.
Neighbor_Net	Good (-65 dBm)	WPA2-Personal	Good signal and secure, but not as strong as Home_WiFi_5G.
Cafe_Free_WiFi	Fair (-78 dBm)	Open Password)	(No Weak signal and no security; not recommended for sensitive activities.

TASK-4

What is SSID Hiding?

SSID hiding is a Wi-Fi feature where the wireless network name (SSID) is **not broadcast** by the router. This means the network name does not appear in the list of available Wi-Fi networks on nearby devices. Users must manually enter the SSID and password to connect.

Advantage of Hiding the SSID

- **Reduces visibility:** The Wi-Fi network is not easily visible to casual users or nearby devices, which may discourage unauthorized people from trying to connect.

Disadvantage of Hiding the SSID

- **Not a strong security measure:** The network can still be detected using Wi-Fi scanning tools, and users must manually enter the network name on every new device, making it less convenient.